# CORE PURE PRACTICE PAPER

# AS-LEVEL

**The total for this paper is 60 marks**

**Advice:**

- **Read every question carefully, highlight/underline key information.**
- **Plan a route to the answer, don't mindlessly calculate, as you might miss something**
- **Make sure your answer is reasonable, can a cat be 50 meters tall?**
- **Attempt every question, the first few marks might not be as hard to achieve**
- **When you check a question, assume you made a mistake, then try to find it**
- **Finish the paper quickly, but accurately, and then quadruple check it. Your second set of eyes is always better than your first**
- **Be clear with your method, minimise the number of arrows or boxes you use.**

**Mark totals:**

**1 – 7 marks, 2 – 6 marks, 3 – 9 marks, 4 – 6 marks, 5 – 4 marks, 6 – 5 marks,**

**7 – 5 marks, 8 – 7 marks, 9 – 11 marks**

**1**

(i) Show that $\frac{1}{x^2-1} = \frac{1}{2x-2} - \frac{1}{2x+2}$ **[2]**

(ii) (A) Find $\sum_{r=2}^{n} \frac{2}{x^2-1}$ **[4]**

(B) Hence, Find $\sum_{r=2}^{\infty} \frac{2}{x^2-1}$ **[1]**

**2** You are given two planes which are perpendicular:

$$\Pi_1: 4x + 5y - 2z = 48$$
$$\Pi_2: ax - 2y + 3z = 74$$

(i) Find the value of the constant $a$ **[3]**

Provided is another plane

$$\Pi_3: 7x + y + 2z = 43$$

(ii) By considering matrices, find the point of intersection of $\Pi_1, \Pi_2 \; and \; \Pi_3$ **[3]**

**3** **For this question you must show detailed reasoning**

Here are two matrices:

$M = \begin{pmatrix} 1 & 4 \\ 6 & 2 \end{pmatrix} \qquad N = \begin{pmatrix} 3 & 1 \\ a & 5 \end{pmatrix}$

(i) (A) A matrix $D$ is equal to $MN$, find D, in terms of $a$ **[2]**

(B) A triangle with coordinates $A(5,0)\ B(12,0)\ C(5,8)$, undergoes the transformation associated with D.

Find the area of the triangle after the transformation, in terms of $a$ **[3]**

(ii) Find any invariant lines of the transformation $M$ that pass through the origin

Leave your answers in exact form **[4]**

**4** **For this question you must show detailed reasoning**

You are given the following complex loci:

$$|z - 4 + 2i| = 5$$
$$\arg(z + i) = \frac{\pi}{4}$$

Find $Im(z)$ **[6]**

**5** **For this question you must show detailed reasoning**

A cubic, with real coefficients, has roots $2 + \sqrt{3}i$, $j$, k.

(i) Explain why $j = 2 - \sqrt{3}i$ **[1]**

(ii) Find an expression for the cubic of the form $z^3 + az^2 + bz + c$, in terms of k **[3]**

**6** Prove $\sum_{r=1}^{n} 3r^2 = \frac{1}{2}n(n+1)(2n+1)$ for all $n \in \mathbb{N}$ **[5]**

**7** The transformation in the xy plane by a matrix $\boldsymbol{M}$ is given by a rotation $30°$ clockwise, about the origin

(i) Determine $\boldsymbol{M}$ and state $\det \boldsymbol{M}$ **[2]**

(ii) State the transformation given by $\boldsymbol{M^{-1}}$, providing its geometrical significance **[3]**

**8** **For this question you must show detailed reasoning**

The polynomial $x^3 + 15 - 4x^2 - 5x$ has roots $\alpha, \beta, \gamma$.

Consider the vector $\mathbf{u} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$ and the vector $\mathbf{v} = 5 \times \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Without finding $\alpha, \beta \; or \; \gamma$, find the acute angle between **u** and **v** to 2 d.p **[7]**

**9** **For this question you must show detailed reasoning**

You are provided the following complex numbers

$$z_1 = 2 - 7i \qquad\qquad z_2 = 5\left(\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right)$$

(i) (A) Find $\frac{1}{z_1}$ **[2]**

(B) Hence, write $\frac{1}{z_1}$ in modulus argument form

Provide the modulus in exact form, and argument to 4 d.p **[3]**

(ii) (A) Sketch $z_2$ and $z_2^*$ on an argand diagram **[1]**

(B) Find $z_2^*$ in modulus argument form **[1]**

(iii) (A) Find $\left|\frac{z_2^*}{z_1}\right|$ in exact form **[2]**

(B) Find $\arg(z_2 \times z_1)$ to 2 d.p **[2]**